

GALOIS DESCENT FOR AFFINE BOST–CONNES BOUNDARIES: CAPITULATION, THE OBSTRUCTION COCYCLE, AND THE EMERGENCE OF THE CLASS GROUP

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ABSTRACT. Let $\mathfrak{p}$ be a non-principal prime of K , $h = \text{ord}([\mathfrak{p}] \in \text{Cl}_K)$, H the Hilbert class field, $G = \text{Gal}(H/K) \cong \text{Cl}_K$. By the principal ideal theorem $\mathfrak{p}\mathcal{O}_H = (\Pi)$ is principal, and $\mathfrak{p}$ splits in H into $g = |\text{Cl}_K|/h$ primes of residue degree h , the Frobenius being the class $[\mathfrak{p}]$ itself under Artin. We construct the capitulated affine system over H on $\prod_{i=1}^g \mathcal{O}_{\mathfrak{P}_i}$, with unit-extended scaling monoid $\mathcal{O}_H^\times \Pi^\mathbb{N}$ — the extension is forced: $\sigma(\Pi) = u_\sigma \Pi$ with $u_\sigma \in \mathcal{O}_H^\times$, so G acts only on the unit-extended system — and prove the descent theorems. First, the obstruction cocycle $c_{\mathfrak{p}} : \sigma \mapsto \sigma(\Pi)/\Pi$ represents $\delta([\mathfrak{p}])$ under a canonical injection $\delta : \text{Cl}_K \hookrightarrow H^1(G, \mathcal{O}_H^\times)$ built from Hilbert 90 and the principal ideal theorem: *the full class group is exactly the group of affine descent obstructions*, and an invariant generator exists if and only if $\mathfrak{p}$ is principal. Second, the three scalings are related by the norm, $\pi = N_{H/K}(\Pi)$ up to $\mathcal{O}_K^\times$, and Green imprimitivity reduces $\partial \mathcal{A}_{H,\Pi}^{\text{aff}} \rtimes G$ to the inert local model crossed by the decomposition group $\langle \text{Frob}_{\mathfrak{p}} \rangle \cong \mathbb{Z}/h$; by Julg this computes the equivariant K-theory. Third, the ladder of unit classes is explicit: $\text{ord}[1] = q^h - 1$ at the bottom ($q = N\mathfrak{p}$; Paper XXXI) and $\text{ord}[1] = q^{|\text{Cl}_K|} - 1$ at the top — the order of the class at the bottom, the class *number* at the top, and the group structure itself in the obstruction cocycles. The integral equivariant K-groups beyond the reduction, and the Morita question for the fixed-point (rather than crossed-product) relation, are recorded as the remaining problems.

1. THE CAPITULATED SYSTEM AND THE GALOIS ACTION

Throughout, $\mathfrak{p}$ is a prime of K with $q = N\mathfrak{p}$ and $h = \text{ord}([\mathfrak{p}] \in \text{Cl}_K)$; H is the Hilbert class field, $G = \text{Gal}(H/K) \cong \text{Cl}_K$ by Artin, and $\mathfrak{p}\mathcal{O}_H = (\Pi)$ by the principal ideal theorem [4, VI.7].

Lemma 1.1 (Splitting structure). *$\mathfrak{p}$ splits in H into $g = |\text{Cl}_K|/h$ distinct primes $\mathfrak{P}_1, \dots, \mathfrak{P}_g$, unramified of residue degree $f = h$, and G permutes them transitively with stabilizer of $\mathfrak{P}_1$ the cyclic group $C := \langle \text{Frob}_{\mathfrak{p}} \rangle \cong \mathbb{Z}/h$. Moreover $N_H(\mathfrak{p}\mathcal{O}_H) = q^{|\text{Cl}_K|}$.*

Proof. H/K is abelian unramified; the Artin map sends $\text{Frob}_{\mathfrak{p}}$ to $[\mathfrak{p}]$, of order h , so $f = h$ and $g = [H : K]/f = |\text{Cl}_K|/h$; the norm of the extended ideal is $q^{fg} = q^{|\text{Cl}_K|}$. $\square$

Proposition 1.2 (The top system, and why the units are forced). *On $X_H := \prod_{i=1}^g \mathcal{O}_{\mathfrak{P}_i}$, with $\mathcal{O}_H$ acting densely by translation (strong approximation) and the monoid $M_H := \mathcal{O}_H^\times \cdot \Pi^\mathbb{N}$ acting by multiplication ($v_{\mathfrak{P}_i}(\Pi) = 1$ for every i), the affine system $\mathcal{A}_{H,\Pi}^{\text{aff}} := C(X_H) \rtimes (\mathcal{O}_H \rtimes M_H)$ and its boundary quotient $\partial \mathcal{A}_{H,\Pi}^{\text{aff}}$ — Cuntz relation over the $q^{|\text{Cl}_K|}$ translates $\mathcal{O}_H/(\Pi)$ — are well defined, and G acts on them by*

$$\beta_\sigma(f) = f \circ \sigma^{-1}, \quad \beta_\sigma(u_b) = u_{\sigma b}, \quad \beta_\sigma(\mu_\gamma) = \mu_{\sigma\gamma} \quad (\gamma \in M_H),$$

using $\sigma(X_H) = X_H$ (permutation of the factors through the $\mathfrak{P}_i$) and $\sigma(M_H) = M_H$. The unit extension is forced: $\sigma(\Pi) = u_\sigma \Pi$ with $u_\sigma = \sigma(\Pi)/\Pi \in \mathcal{O}_H^\times$ (both sides generate the G -invariant ideal $\mathfrak{p}\mathcal{O}_H$), so on the minimal monoid $\Pi^\mathbb{N}$ the Galois action is defined only up to the unit twist

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Paper XXXII of the series on localized Bost–Connes systems. The framework is that of [1], cited by DOI; Papers XXX and XXXI of the series are referred to by numeral, and every fact used from them is restated in place.

u_σ , and no sub-monoid on which G acts omits the units unless the twist is a coboundary — which, by Theorem 2.1, happens exactly when $\mathfrak{p}$ is principal.

2. THE OBSTRUCTION COCYCLE IS THE CLASS

Theorem 2.1 (Emergence of the full class group). *The map $\sigma \mapsto u_\sigma = \sigma(\Pi)/\Pi$ is a 1-cocycle whose class $[c_{\mathfrak{p}}] \in H^1(G, \mathcal{O}_H^\times)$ is independent of the choice of Π , and there is a canonical injective homomorphism*

$$\delta : \text{Cl}_K \hookrightarrow H^1(G, \mathcal{O}_H^\times), \quad \delta([\mathfrak{a}]) = [\sigma \mapsto \sigma(\alpha)/\alpha] \text{ for } \mathfrak{a}\mathcal{O}_H = (\alpha),$$

with $\delta([\mathfrak{p}]) = [c_{\mathfrak{p}}]$. In particular: an invariant generator of $\mathfrak{p}\mathcal{O}_H$ exists if and only if $\mathfrak{p}$ is principal in K ; and the group of descent obstructions of the affine systems, as $\mathfrak{p}$ ranges over the primes of K , is exactly $\delta(\text{Cl}_K) \cong \text{Cl}_K$.

Proof. Cocycle identity and independence of Π are immediate ($\Pi \mapsto v\Pi$ changes $c_{\mathfrak{p}}$ by the coboundary of v). For δ : every ideal capitulates ($\mathfrak{a}\mathcal{O}_H$ principal, the principal ideal theorem [4, VI.7]), so δ is defined on all of Cl_K ; it is well defined and homomorphic by the same coboundary computation, principal ideals of K mapping to coboundaries of elements of $K^\times$, which are trivial. Injectivity: if $\sigma(\alpha)/\alpha = \sigma(w)/w$ for all σ and some $w \in \mathcal{O}_H^\times$, then $\alpha/w \in H^{\times G} = K^\times$ and $\mathfrak{a}\mathcal{O}_H = (\alpha/w)\mathcal{O}_H$ with $\alpha/w \in K^\times$; taking valuations at the primes of K (unramifiedness) gives $\mathfrak{a} = (\alpha/w)$ in $\mathcal{O}_K$, so $[\mathfrak{a}] = 1$. (Equivalently: δ is the connecting map of $1 \rightarrow \mathcal{O}_H^\times \rightarrow H^\times \rightarrow P_H \rightarrow 1$ restricted to extended ideals, injective by Hilbert 90.) The last statement holds since prime classes generate Cl_K . $\square$

3. DESCENT: NORM, IMPRIMITIVITY, AND THE LADDER OF UNIT CLASSES

Proposition 3.1 (The bottom scaling is the norm of the top scaling). *$(N_{H/K}(\Pi)) = \mathfrak{p}^{|\text{Cl}_K|} = (\pi)^{|\text{Cl}_K|/h}$, so $N_{H/K}(\Pi) = \pi^g u$ for a unit $u \in \mathcal{O}_K^\times$: the Paper XXXI scaling π is, up to units and the transitivity index g , the norm of the capitulated scaling. For $g = 1$ (that is, $h = |\text{Cl}_K|$, the class $[\mathfrak{p}]$ generating), $\pi = N_{H/K}(\Pi)u$ exactly.*

Proof. $N_{H/K}(\mathfrak{p}\mathcal{O}_H) = \mathfrak{p}^{[H:K]} = \mathfrak{p}^{hg} = (\pi)^g$; apply $N_{H/K}$ to Π . $\square$

Theorem 3.2 (Imprimitivity, and the equivariant K-theory reduced). *By Julg, $K_*^G(\partial\mathcal{A}_{H,\Pi}^{\text{aff}}) \cong K_*(\partial\mathcal{A}_{H,\Pi}^{\text{aff}} \rtimes G)$ [3]; and since $X_H \cong \text{Ind}_G^G \mathcal{O}_{\mathfrak{p}_1}$ as a G -space (Lemma 1.1), Green's imprimitivity theorem [2] gives a strong Morita equivalence*

$$\partial\mathcal{A}_{H,\Pi}^{\text{aff}} \rtimes G \sim_M \partial\mathcal{A}_{\mathfrak{p}_1}^{\text{aff,loc}} \rtimes C,$$

where $\partial\mathcal{A}_{\mathfrak{p}_1}^{\text{aff,loc}}$ is the inert local model — the boundary system on $\mathcal{O}_{\mathfrak{p}_1}$ alone, translations by $\mathcal{O}_H$ through its dense image, scalings by $\mathcal{O}_H^\times \Pi^\mathbb{N}$ — and $C \cong \mathbb{Z}/h$ is the decomposition group, acting through the local Frobenius. The computation of the integral $K_*(\partial\mathcal{A}_{\mathfrak{p}_1}^{\text{aff,loc}} \rtimes \mathbb{Z}/h)$ is the reduction achieved; it is not completed here (Problem (i) of §4).

Theorem 3.3 (The ladder of unit classes). *In K_0 of the respective boundary quotients:*

$$\text{ord}[1_{\partial\mathcal{A}_{K,\{\mathfrak{p}\}}^{\text{aff}}}] = q^h - 1 \quad \text{and} \quad \text{ord}[1_{\partial\mathcal{A}_{H,\Pi}^{\text{aff}}}] = q^{|\text{Cl}_K|} - 1.$$

The bottom sees the order of the class, the top the class number, and the group structure of Cl_K itself lives in the obstruction cocycles of Theorem 2.1.

Proof. The bottom is Paper XXXI. For the top, the $r = 0$ layer of the exterior-tower argument of Paper XXX applies on X_H : $C(X_H) = \varinjlim_m C(\mathcal{O}_H/(\Pi^m))$ with transitive translation action, $|\mathcal{O}_H/(\Pi)| = q^{|\text{Cl}_K|}$ by Lemma 1.1, so $M_0 = \mathbb{Z}[1/q]$ with the scaling acting by $q^{-|\text{Cl}_K|}$, and

$\text{coker}(1 - q^{-|\text{Cl}_K|}) = \mathbb{Z}/(q^{|\text{Cl}_K|} - 1)$ generated by the class of the unit; the additional unit scalings act trivially on M_0 (they preserve every Π -adic cylinder's measure class and fix the class of 1), so the order is exact. $\square$

4. WHAT REMAINS

(i) The integral computation of $K_*(\partial\mathcal{A}_{\mathfrak{p}_1}^{\text{aff,loc}} \rtimes \mathbb{Z}/h)$: the exterior tower of Paper XXX is $\mathbb{Z}/h$ -equivariant, and what is needed is the equivariant colimit — concretely, the $R(\mathbb{Z}/h)$ -module structure of the layers $M_r(\mathcal{O}_H)$ under Frobenius — not a new idea but a genuine computation, already nontrivial for the first inert example ($K = \mathbb{Q}(\sqrt{-5})$, $H = \mathbb{Q}(i, \sqrt{5})$, $\mathfrak{p}_3$ inert in H/K , $C = G = \mathbb{Z}/2$). (ii) The fixed-point question: the Paper XXXI bottom system is the C -fixed-point system of the inert local model ($\mathcal{O}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}_1}^C$, $\mathcal{O}_K = \mathcal{O}_H^C$, $\pi = N(\Pi)u$), and whether fixed-point and crossed-product descents are Morita equivalent is governed by the properness of the C -action on the boundary groupoid — measure-theoretically the fixed locus is null, topologically it is not, and the discrepancy is exactly where the obstruction cocycle $[c_{\mathfrak{p}}]$ should reappear. (iii) The unit-extended monoid enlarges the top algebra by the infinite unit group $\mathcal{O}_H^\times$; its own K-theoretic footprint (a torus of rank the unit rank of H) must be separated from the class-group data in any complete answer.

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